

Closed-Loop Φ -Based Intervention During Quantum Circuit Execution on IBM Quantum Hardware

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May 2026

Abstract

We investigate whether a training-free thermodynamic stability metric $\Phi = I \times \rho - \alpha \times S$, computed from IBM Quantum calibration or telemetry data, can drive closed-loop intervention decisions during execution-associated monitoring windows on real hardware. Across 10 paid circuit tests on three IBM backends (ibm_fez, ibm_marrakesh, ibm_torino), together with simulated and telemetry-based harness validation, we evaluate a closed-loop controller that monitors Φ during an execution-associated window and supports five intervention actions: CONTINUE, CONTINUE-DEGRADED, CLASSICAL-FALLBACK, CHECKPOINT-MIGRATE, and ABORT-RESTART. In a same-circuit controlled comparison on ibm_fez, swapping LOW- Φ qubits ($\Phi = 0.097, 0.030$) for HIGH- Φ qubits ($\Phi = 0.9988, 0.9984$) reduced error from 18.36% to 2.73% on an identical Deutsch-Jozsa circuit, an 85.1% relative error reduction. Under matched conditions, composite Φ selection outperformed raw T_2 selection by 7.79 percentage points on ibm_fez (68.7% relative reduction) and by 6.37 percentage points aggregate across three backends, with ibm_marrakesh as a documented counterexample where T_2 selection won despite lower Φ on the winning pair. Mid-circuit dynamic branching on IBM hardware achieved 98.78% per-shot conditional consistency between probe outcome and branch marker, with an estimated 101,220 logical gates avoided on the ABORT path. A Wilson confidence interval non-overlap rule triggered the ABORT-RESTART action on 1 of 3 backends in the tested setting, correctly holding on the other two where confidence intervals overlapped. Classical checkpoint representation at a segmentation boundary enabled suffix migration from LOW- Φ to HIGH- Φ qubits with 82.15% final fidelity, without storing an unknown quantum state. Honest counterexamples are preserved: in a three-policy cost comparison a seeded random pair outperformed the Φ -guided pair on ibm_fez, and on ibm_marrakesh raw T_2 selection outperformed Φ selection under matched compiled circuit conditions. These results support a closed-loop intervention framework in which Φ can operate as a control variable during execution-associated monitoring rather than only as a pre-execution selection score, with intervention actions selected by a priority-based decision hierarchy and bounded by confirmation-window, cooldown, and maximum-intervention guardrails.

1 Introduction

Quantum computing systems execute quantum circuits on physical qubits that are subject to decoherence, noise, drift, and other sources of performance variation. In many current workflows, execution is open-loop: qubits are selected before execution, the circuit is submitted, and results are returned after completion, with limited ability to respond to changing conditions while the computation is in progress. When qubit quality degrades during execution, or when pre-execution selection relied on calibration data that no longer reflects current hardware state, the shot budget, queue time, and hardware access time spent on that circuit are partially or entirely wasted. Earlier detection of degradation could enable actions that reduce waste, preserve partial progress, or redirect remaining computation to a different execution mode.

Prior work using the thermodynamic stability metric $\Phi = I \times \rho - \alpha \times S$ has focused on pre-execution decisions. In earlier work on qubit selection, Φ computed from IBM Quantum calibration data was shown to identify high-quality qubits before circuit execution, reducing error by approximately 83% in a qubit-selection test on `ibm_fez` (Barnicle, 2026a). In sibling work on quantum versus classical routing, Φ was shown to identify a crossover regime in the tested setting where LOW- Φ quantum execution produced higher error than exact classical simulation on classically tractable circuits, supporting pre-execution allocation decisions between quantum hardware and classical fallback (Barnicle, 2026b). These works establish Φ as a pre-execution selection and routing signal. The question addressed in the present paper is different: can Φ operate as a closed-loop control variable during execution-associated monitoring windows, driving intervention actions while the workflow remains active, queued, segmented, checkpointed, or resumable?

Closing the loop during execution raises four questions that pre-execution methods do not answer. First, can a controller ingest Φ from live backend telemetry and use it to choose among intervention actions in a way that measurably changes realized hardware outcomes? Second, can intervention be triggered by statistical evidence of separation between candidate qubit pools rather than by hardcoded fidelity thresholds, which are brittle and ignore measurement uncertainty? Third, can computational progress be preserved when LOW- Φ qubits are detected after partial execution, without violating the no-cloning constraint on unknown quantum states? Fourth, can true mid-circuit intervention, in which a classical conditional branch is taken based on an in-circuit measurement, be demonstrated on current IBM dynamic circuits rather than only in simulation?

The evidence presented in this paper supports affirmative answers to these four questions in the tested setting. The central finding is that Φ can operate as a closed-loop control variable during execution-associated monitoring windows, with intervention actions selected by a priority-based decision hierarchy and bounded by confirmation-window, cooldown, and maximum-intervention guardrails. Two layers of evidence support this claim. A strict hardware layer of 10 paid circuit tests on three IBM backends validates CONTINUE, CHECKPOINT-MIGRATE, ABORT-RESTART, mid-circuit dynamic branching, and statistical intervention. A supporting layer of simulated harness and telemetry-based calibration tests exercises the full five-action decision hierarchy including CLASSICAL-FALLBACK and CONTINUE-DEGRADED, which have stronger evidence at the harness layer than at the paid hardware layer. Honest counterexamples are preserved rather than hidden: on `ibm_marrakesh` under matched compiled circuit conditions, raw T_2 selection outperformed Φ selection despite lower Φ on the winning pair,

and in a three-policy cost comparison on `ibm_fez` a seeded random pair outperformed the Φ -guided pair.

The main contributions of this paper are:

1. Empirical evidence that closed-loop Φ -driven intervention during execution-associated monitoring windows can improve realized hardware outcomes on current IBM Quantum hardware, including a controller-authorized 85.1% relative error reduction from LOW- Φ to HIGH- Φ qubits on an identical Deutsch-Jozsa circuit and a 68.7% relative error reduction from composite Φ selection over raw T_2 selection under matched conditions on `ibm_fez`.
2. A Wilson confidence interval non-overlap rule that triggers ABORT-RESTART based on statistical evidence of separation between probe runs, tested on three IBM backends and correctly intervening on 1 of 3 where confidence intervals did not overlap.
3. A classical checkpoint representation at a segmentation boundary that enables suffix migration from LOW- Φ to HIGH- Φ qubits without violating the no-cloning constraint on unknown quantum states, with 82.15% final fidelity in the tested migration.
4. Hardware evidence for true mid-circuit intervention using IBM dynamic circuits, with 98.78% per-shot conditional consistency between probe outcome and branch marker across 4096 shots.

2 Related Work

Quantum hardware quality assessment and qubit selection have been studied primarily through calibration metrics and post-compilation routing. Nation and Treinish (Nation and Treinish, 2023) developed `mapomatic`, a post-compilation layout optimizer that scores qubit subsets using heuristic cost functions derived from calibration data and recovers approximately 40% fidelity in their tested setting. Their analysis did not explicitly model qubit idle-time errors, reporting that such terms did not greatly change layout ordering because contributions from T_1 and T_2 were already partly reflected in instruction errors (Nation and Treinish, 2023). `Mapomatic` operates post-compilation and selects the best available hardware assignment among quantum options. The present work operates during execution-associated monitoring windows and answers a different question: given that a circuit is already executing or queued on hardware, what intervention action should be taken if stability degrades or if telemetry indicates a better execution target has become available? Sibling work using the same Φ formulation addressed the pre-execution quantum-versus-classical routing decision (Barnicle, 2026b). The present paper does not replace either approach; it adds the during-execution control layer that sits between pre-execution routing and post-execution analysis.

Quantum error mitigation and quantum error correction are distinct in purpose from the present work. Error mitigation methods such as zero-noise extrapolation (Temme et al., 2017) operate during or after execution and reduce bias in measured expectation values through post-processing or noise extrapolation. Quantum error correction uses syndrome measurements to detect and correct errors on logical information encoded redundantly across physical qubits. Neither framework addresses the execution-management

decision of whether to continue a circuit, migrate to better qubits, fall back to classical simulation, or restart on a different configuration based on a stability metric. The present work is execution-management control, not error mitigation or correction, and the two can be composed: mitigation or correction may operate within a CONTINUE action, while the controller selects among intervention actions at the workflow level.

Mid-circuit measurement and dynamic circuits on IBM hardware have been used in settings such as quantum error correction and adaptive quantum protocols, including measurement-based computation. The present work uses mid-circuit measurement for a different purpose: sensing execution-state stability and driving a classical conditional branch that selects between CONTINUE and ABORT intervention paths. Variational quantum algorithms (Cerezo et al., 2021) hybridize quantum and classical computation at the algorithm level, decomposing problems so that parameterized quantum circuits are optimized by a classical outer loop. The present work hybridizes at the control layer rather than the algorithm layer: the decision of whether to execute, continue, migrate, or fall back is a classical control decision driven by a stability signal, regardless of whether the quantum circuit is itself variational.

3 Methods

3.1 The Phi Stability Metric

The closed-loop controller uses the thermodynamic stability metric:

$$\Phi = I \times \rho - \alpha \times S \quad (1)$$

where the components are computed from IBM Quantum calibration or telemetry data as follows. Normalized fidelity is $I = (F - 0.50)/0.50$, clamped to $[0, 1]$, where $F = 1 - \text{single-qubit gate error}$ is single-qubit gate fidelity and 0.50 is the reference baseline for two-level systems. Coherence ratio is $\rho = \min(T_2/T_1, 1.0)$, where T_1 is relaxation time and T_2 is dephasing time. Readout error is $S = E$, the calibration-reported readout error probability, clamped to $[0, 1]$. The coupling constant is $\alpha = 0.1$ and the intervention threshold is $\Phi_c = 0.25$.

For circuits using multiple qubits, the controller evaluates the minimum Φ across the set of qubits participating in the circuit:

$$\Phi_{\min} = \min_{i \in \text{circuit qubits}} \Phi_i \quad (2)$$

The minimum is used as a conservative weakest-link criterion. Other aggregation rules (mean, median, weighted, percentile-based, or critical-subset rules) are permissible in the framework but were not used in the tested experiments.

3.2 IBM Quantum Hardware and Calibration Data

Calibration data and circuit execution used three IBM Quantum backends: `ibm_fez` (156 qubits), `ibm_torino` (133 qubits), and `ibm_marrakesh` (156 qubits). For each qubit, the calibration record contains single-qubit gate fidelity F , relaxation time T_1 , dephasing time T_2 , and readout error E . Φ was computed from these calibration fields for each qubit before each experiment, and, where applicable, re-queried at segmentation boundaries within a multi-segment test. All circuits and transpilation steps were implemented in

Qiskit, hardware execution used IBM Quantum Runtime, and dynamic-circuit executions used IBM Sampler primitives with mid-circuit measurement enabled. No synthetic data were used in the telemetry-based or paid hardware experiments. A separate simulated harness layer used synthetic Φ sequences solely to verify decision logic and guardrails.

3.3 Five-Action Intervention Framework

The framework provides five intervention actions. The paid hardware tests in this paper exercised CONTINUE, CHECKPOINT-MIGRATE, pre-submit ABORT-RESTART, and statistical ABORT-RESTART on real hardware. CLASSICAL-FALLBACK and CONTINUE-DEGRADED were exercised at the simulated harness layer only. The framework organizes the five actions as a priority-based decision hierarchy evaluated in order:

1. **CONTINUE.** Execution proceeds without intervention. Selected when $\Phi_{\min} \geq \Phi_c$ and no intervention criterion is satisfied.
2. **CONTINUE-DEGRADED.** Execution proceeds despite degraded stability. Selected when execution is near completion or when alternative options are unavailable or not cost-effective.
3. **CLASSICAL-FALLBACK.** Remaining computation is redirected to classical simulation. Selected when remaining computation is determined to be classically tractable based on remaining qubit count, remaining gate count, and deployment-specific tractability bounds.
4. **CHECKPOINT-MIGRATE.** A classical checkpoint representation is preserved at a segmentation boundary and the remaining computation resumes on alternative qubits. Selected when execution progress satisfies a migration-eligibility condition and migration targets are available. The checkpoint explicitly excludes storage of an unknown quantum state; it stores a classical representation of a measured outcome at a segmentation boundary, which is sufficient to resume a parameterized suffix computation.
5. **ABORT-RESTART.** Current execution is terminated and the circuit is restarted using different qubits. Selected when execution is in an early stage or when restarting is more cost-effective than continuing. Supports both a pre-submit variant (the aborted attempt never reaches hardware) and a statistical variant (triggered by probe-based evidence of separation, described in the next subsection).

In the tested controller configuration, the near-completion condition for CONTINUE-DEGRADED was set to progress $\geq 85\%$, the CLASSICAL-FALLBACK tractability bounds were set to remaining qubits ≤ 25 and remaining gates ≤ 1000 , and the migration-eligibility condition for CHECKPOINT-MIGRATE was set to progress $\geq 50\%$. These values correspond to a conservative estimate of classical-simulation tractability on commodity hardware and are configurable by deployment.

3.4 Guardrails

The controller includes guardrails that reduce excessive or unstable intervention behavior. A confirmation requirement delays intervention selection until $\Phi_{\min}$ has satisfied

the below-threshold condition for a specified number of consecutive evaluation cycles. A cooldown period suppresses further interventions for a specified number of evaluation cycles after an action is selected. A maximum-intervention count limits the total number of interventions per circuit execution. In the tested controller configuration, the confirmation requirement was set to 2 cycles, the cooldown period was set to 3 cycles, and the maximum-intervention count was set to 5 per circuit execution. These defaults are configurable and may be adjusted based on deployment characteristics.

3.5 Wilson Confidence Interval Intervention Rule

A statistical intervention rule compares Wilson confidence intervals between a candidate baseline pool (typically selected by raw T_2) and a candidate Φ pool. For each pool, a short probe run of 256 shots is executed, and a Wilson confidence interval at the 95% level is computed for the probe fidelity. If the lower bound of the Φ -pool interval strictly exceeds the upper bound of the baseline-pool interval, the controller triggers ABORT-RESTART on the baseline path and commits the full remaining shot budget to the Φ pool. If the intervals overlap, a continue-baseline tie policy applies and no intervention is triggered. This rule replaces hardcoded fidelity thresholds with a statistical separation criterion that explicitly accounts for measurement uncertainty, at the cost of probe overhead.

3.6 Validation Protocol

The closed-loop controller was validated across three layers. The simulated harness layer exercised the four non-CONTINUE intervention actions using synthetic Φ sequences at $\Phi = 0.19$ across varying progress levels, confirming that the decision hierarchy selects the correct action for each (progress, resource) combination. The telemetry-based harness layer connected to live IBM backends, retrieved calibration data for available qubits, computed Φ , and verified that the controller correctly ingested live telemetry, applied caching and throttling (1 API call, 5 cached), and triggered interventions under injected degradation without executing circuits. The paid hardware layer consisted of 10 real circuit-execution tests on three IBM backends between January 15 and January 29, 2026, using approximately 228 seconds of IBM Quantum Runtime credits in total. Each paid test produced a structured audit record containing the run identifier, backend, selected qubits, Φ values, intervention action, decision parameters, job identifier, and execution counts. Structured results summaries, repository-tracked documentation, and code are available in the repository identified at the end of this paper. Audit records are generated by the validated test suite and described in the repository documentation.

Paid test circuits were drawn from the Deutsch-Jozsa family (2-qubit balanced, used in Tests 1, 2, 8, 9, 10), repeated Deutsch-Jozsa units of varying depth (Tests 3, 4, 5, 6), and a dynamic-circuit design using a sentinel qubit (Test 7). Expected outcomes were determined analytically from the circuit structure prior to execution. Fidelity was computed as the fraction of shots returning the expected bitstring; error was computed as $1 - \text{fidelity}$. Shot counts were 4096 for full runs and 256 for probe runs in the Wilson CI protocol, unless otherwise stated in the individual test descriptions.

4 Results

4.1 Primary Same-Circuit Controlled Comparison: LOW- Φ vs HIGH- Φ Qubits on `ibm_fez`

The primary controlled comparison used an identical Deutsch-Jozsa balanced circuit on `ibm_fez`, varying only the qubit pair selected for execution. Test 1 (January 15, 2026) executed the circuit on a LOW- Φ pair [149, 150] with Φ values 0.0970 and 0.0300. Test 2 (January 17, 2026) executed the same circuit on a controller-selected HIGH- Φ pair [138, 151] with Φ values 0.9988 and 0.9984. Each test used 4096 shots, with the expected output being $|01\rangle$ for the balanced Deutsch-Jozsa circuit.

Table 1: Primary Controlled Comparison: LOW- Φ vs HIGH- Φ on Identical Deutsch-Jozsa Circuit (`ibm_fez`)

Qubit Pair	Φ Values	Error (%)
LOW- Φ [149, 150]	0.0970, 0.0300	18.36
HIGH- Φ [138, 151]	0.9988, 0.9984	2.73

The LOW- Φ execution produced 18.36% error (3344 shots on $|01\rangle$, 673 on $|00\rangle$, 60 on $|11\rangle$, 19 on $|10\rangle$). The HIGH- Φ execution produced 2.73% error (3984 shots on $|01\rangle$, 66 on $|00\rangle$, 25 on $|11\rangle$, 21 on $|10\rangle$). The absolute reduction was 15.63 percentage points, corresponding to an 85.1% relative error reduction. The circuit, backend, shot count, and analysis protocol were the same across the two runs, with qubit-pair choice as the primary difference under comparison.

4.2 Composite Φ vs Raw T_2 Under Matched Conditions

Test 8 (January 29, 2026) compared composite Φ selection against raw T_2 selection on `ibm_fez` under matched conditions. Both policies executed the same Deutsch-Jozsa balanced circuit at 4096 shots on the same day. The Φ policy selected qubits [151, 152] on composite stability; the RAW policy selected qubits [87, 88] by T_2 alone (T_2 scores 0.000206 and 0.000239).

Table 2: Composite Φ vs Raw T_2 Selection on `ibm_fez`, Same Circuit

Policy	Qubit Pair	Error (%)
Composite Φ	[151, 152]	3.54
Raw T_2	[87, 88]	11.33

Composite Φ produced 3.54% error; raw T_2 produced 11.33% error. The absolute difference was 7.79 percentage points, corresponding to a 68.7% relative error reduction for composite Φ over the single-field T_2 baseline in the tested setting. This result addresses the "why not just pick qubits by T_2 ?" question directly: the composite metric added measurable value over the strongest single-field calibration alternative in this same-circuit same-day comparison.

4.3 Cross-Backend Generalization with Documented Counterexample

Test 9 (January 29, 2026) replicated the Test 8 protocol across three IBM backends under matched compiled circuit conditions (compiled depth 16, two-qubit gate count 1, zero swaps). The outcome was logged regardless of which policy won on each backend. Test 9 is an independent run replicating the Test 8 protocol; the `ibm.fez` numbers differ from Test 8 by approximately 0.05 to 0.29 percentage points due to run-to-run variation in a separate sampling draw on the same qubit pairs.

Table 3: Cross-Backend Φ vs Raw T_2 Comparison (Deutsch-Jozsa, 4096 Shots Each)

Backend	Φ Pair	Φ Err (%)	RAW Pair	RAW Err (%)	Δ (pp)
<code>ibm.fez</code>	[151, 152]	3.49	[87, 88]	11.04	+7.54
<code>ibm.torino</code>	[12, 18]	4.49	[12, 13]	7.79	+3.30
<code>ibm.marrakesh</code>	[58, 71]	7.32	[53, 54]	2.86	-4.47

The Δ column shows the percentage-point difference between RAW error and Φ error, with positive values favoring Φ . Composite Φ outperformed raw T_2 on 2 of 3 backends. The aggregate net advantage across the three backends was 6.37 percentage points in favor of Φ . The `ibm.marrakesh` result is a documented counterexample: the RAW-selected pair had lower minimum Φ (0.9765) than the Φ -selected pair (0.9976) but still achieved lower realized error under matched compiled circuit conditions. Inspection of the saved execution artifacts found no evidence of a software bug, calibration drift, or compiled-circuit mismatch between the two executions. In this execution interval, calibration-proxy Φ did not dominate realized sampling error, supporting a bounded claim of majority-backend advantage rather than uniform dominance.

4.4 Mid-Circuit Dynamic Branching with Marker-Verified Consistency

Test 7 (January 29, 2026) implemented true mid-circuit intervention using IBM dynamic circuits. Compute qubits [0, 1] executed a 12-block cancel-work structure, with a sentinel qubit [2] probed at block 6. A marker bit on the compute qubits was set conditionally on the probe outcome: `probe = 0` drove the CONTINUE branch and set the marker to 1; `probe = 1` drove the ABORT branch and set the marker to 0. Per-shot verification was obtained by comparing the probe measurement against the marker bit across 4096 shots.

Table 4: Mid-Circuit Branching on `ibm.fez`: Joint Probe and Marker Outcomes

Joint Outcome	Shots	Interpretation
<code>probe=0, marker=1</code>	2385	CONTINUE branch, consistent
<code>probe=1, marker=0</code>	1661	ABORT branch, consistent
<code>probe=0, marker=0</code>	26	Inconsistent
<code>probe=1, marker=1</code>	24	Inconsistent

Counting by realized branch marker, the CONTINUE branch totaled 2409 shots (2385 consistent plus 24 inconsistent), or 58.8%, and the ABORT branch totaled 1687 shots

(1661 consistent plus 26 inconsistent), or 41.2%. Joint probe-marker consistency was 4046 out of 4096 shots, or 98.78% per-shot conditional consistency. An estimated 101,220 logical gates were avoided on the ABORT path, computed as 6 remaining blocks after the probe, 10 logical gates per block, and 1687 ABORT shots. This test was designed to verify mid-circuit branching mechanics rather than to maximize algorithmic fidelity; the 25.27% overall fidelity reflects the intentional use of heavy cancel-work, dynamic control, and sentinel delay required to produce distinguishable branch behavior.

4.5 Wilson Confidence Interval Intervention Across Three Backends

Test 10 (January 29, 2026) applied the Wilson CI intervention rule to RAW versus Φ probe runs on three IBM backends. Each probe used 256 shots; each full run used 4096 shots. Wilson confidence intervals were computed at the 95% level, with a continue-baseline tie policy applied when intervals overlapped.

Table 5: Wilson CI Intervention Rule Applied to Three IBM Backends

Backend	RAW Probe CI (%)	Φ Probe CI (%)	Decision
ibm_fez	(79.85, 88.65)	(92.47, 97.58)	ABORT-RESTART
ibm_marrakesh	(93.45, 98.14)	(94.98, 98.92)	CONTINUE (RAW)
ibm_torino	(89.62, 95.81)	(95.51, 99.16)	CONTINUE (RAW)

On `ibm_fez`, the RAW probe achieved 84.77% fidelity with CI (79.85%, 88.65%) and the Φ probe achieved 95.70% fidelity with CI (92.47%, 97.58%). The intervals did not overlap, with a separation of 3.82 percentage points between the Φ lower bound (92.47%) and the RAW upper bound (88.65%). The controller triggered ABORT-RESTART on the RAW path and committed the full 4096-shot budget to the Φ pool, which achieved 95.85% final fidelity. On `ibm_marrakesh` and `ibm_torino`, the intervals overlapped (separation of -3.16 pp and -0.30 pp respectively), and the controller correctly applied the continue-baseline tie policy. Full RAW runs on `ibm_marrakesh` and `ibm_torino` achieved 97.46% and 94.70% fidelity respectively. Intervention triggered on 1 of 3 backends in the tested setting, which is the correct behavior when statistical separation is backend-dependent and the controller is designed to intervene only when evidence justifies the cost of the intervention itself.

4.6 Classical Checkpoint Migration to Alternative Qubits

Test 4 (January 29, 2026) exercised the CHECKPOINT-MIGRATE intervention pathway on `ibm_fez`. A 25-repeat Deutsch-Jozsa workload was segmented into a 3-repeat prefix and a 22-repeat suffix at a classically-measurable boundary. The prefix executed on a LOW- Φ pair [8, 9] with minimum Φ 0.0990. After observing the prefix outcome, the controller evaluated the $\Phi_{\min}$ threshold condition (0.0990 below the 0.25 threshold), selected CHECKPOINT-MIGRATE, and migrated the suffix to a HIGH- Φ pair [151, 152] with minimum Φ 0.9984. Each stage used 2048 shots.

Table 6: Checkpoint-Migrate on `ibm_fez`: Prefix-to-Suffix Transition

Stage	Qubit Pair	Min Φ	Observed Fidelity (%)
Prefix (3 repeats)	[8, 9]	0.0990	92.58
Suffix (22 repeats)	[151, 152]	0.9984	82.15

The prefix produced $|01\rangle$ at 92.58% confidence. The controller stored this classical bitstring as the checkpoint representation at the segmentation boundary, parameterized the suffix circuit on the observed outcome, and executed the suffix on the HIGH- Φ pair. The suffix produced $|01\rangle$ at 82.15% fidelity. The checkpoint representation is a classical bitstring corresponding to a measured outcome, not a stored quantum state; the no-cloning constraint is not violated because no unknown quantum state is preserved across the segmentation boundary. A cost proxy computed as compiled depth multiplied by shots gave a delta of 92,160 depth $\times$ shots units between the segmented execution (prefix on LOW- Φ plus suffix on HIGH- Φ) and the counterfactual full restart on HIGH- Φ from the beginning, with the segmented path costing less. Suffix fidelity of 82.15% is lower than the 97.27% achieved by a HIGH- Φ pair on the shorter Test 2 circuit, reflecting the accumulated cost of segmentation plus resume; the result is bounded as "preserves useful output" rather than "lossless."

4.7 Pre-Submit ABORT-RESTART on Extremely Low- Φ Qubits

Test 5 (January 29, 2026) exercised the pre-submit variant of ABORT-RESTART on `ibm_fez`. A 25-repeat Deutsch-Jozsa circuit (compiled depth 376, 2048 shots) was initially targeted at a candidate pair [72, 73] with minimum Φ of -0.0487 . The controller evaluated $\Phi_{\min}$ against the 0.25 threshold before API submission, selected ABORT-RESTART, and redirected execution to a HIGH- Φ pair [123, 124] with minimum Φ 0.9986.

Table 7: Pre-Submit ABORT-RESTART on `ibm_fez`: Avoided Attempt and Restart Outcome

Stage	Qubit Pair	Min Φ	Outcome
Candidate (avoided)	[72, 73]	-0.0487	Not submitted
Restart	[123, 124]	0.9986	70.29% fidelity

The aborted attempt never reached hardware; 1 job and 2048 shots were avoided at zero hardware-execution cost for that attempt because the decision was made entirely from calibration telemetry prior to API submission. The restart on the HIGH- Φ pair achieved 70.29% fidelity (29.71% error). The experiment does not measure the realized fidelity that the avoided pair would have produced, so the supported claim is preventive resource saving rather than a measured completed-run delta; substantially degraded fidelity on a pair with $\Phi = -0.0487$ would be expected from the broader hardware evidence, but it was not measured in this test.

4.8 Marginal- Φ Qubits on Deep Circuits with Flat Calibration

Test 3 (January 29, 2026) characterized the behavior of marginal- Φ qubits on deep circuits while the calibration- derived Φ signal remained stable across the observation window. Qubits [7, 17] had Φ values 0.3747 and 0.7574, with a minimum of 0.3747. The circuit was a 250-repeat Deutsch-Jozsa unit with compiled depth 3751, executed in three segments of 2048 shots each, separated by 40-second sleeps with calibration snapshots taken between segments.

Table 8: Segmented Deep-Circuit Execution on `ibm_fez`: Marginal- Φ Pair Across Three Segments

Segment	Minimum Φ	Fidelity (%)	Error (%)
1	0.3747	33.28	66.72
2	0.3747	30.44	69.56
3	0.3747	30.05	69.95
Aggregate	0.3747	31.26	68.74

Fidelity remained near 30% across all three segments, with aggregate error of 68.74%. The Φ drift across the three calibration snapshots was 0.0: the minimum Φ value reported by the backend was identical at 0.3747 in every snapshot. Depth-induced degradation of this kind was not detected by the periodic calibration snapshots used in this test. The experiment does not isolate mid-execution Φ collapse from depth- induced degradation on marginal qubits, because the Φ signal remained stable across the observation window. The result bounds to "realized depth degradation despite stable calibration snapshots" rather than "observed Φ collapse during execution." The 31.26% fidelity reported here is consistent with circuit depth acting on marginal- Φ qubits under conditions that the periodic calibration snapshots used here did not surface as deteriorating. The experiment does not establish what fidelity a HIGH- Φ pair would have achieved at the same compiled depth of 3751.

4.9 Documented Counterexamples to Φ -Based Selection

Two counterexamples were documented in the paid hardware suite in which a lower- Φ pair outperformed a higher- Φ pair under matched conditions. These are preserved in the scientific record as bounded findings about the calibration proxy.

The first counterexample is from Test 6 on `ibm_fez` (January 29, 2026), a three-policy cost comparison on a 25-repeat Deutsch-Jozsa circuit (compiled depth 376, 2048 shots). A seeded random policy selected pair [64, 65] with minimum Φ 0.7109. A Φ -guided policy selected pair [123, 124] with minimum Φ 0.9986. A Φ -controller policy operating on the same seeded starting pair [64, 65] evaluated $\Phi_{\min}$ at 0.7109 (above the 0.25 threshold), selected NO-ABORT, and submitted the starting pair.

Table 9: Three-Policy Cost Comparison on `ibm_fez`: Random Outperforms Φ -Guided

Policy	Qubit Pair	Min Φ	Fidelity (%)
Random baseline	[64, 65]	0.7109	70.58
Φ -guided	[123, 124]	0.9986	64.31
Φ -controller	[64, 65]	0.7109	69.12

The seeded random pair achieved 70.58% fidelity, outperforming the Φ -guided pair at 64.31% fidelity despite the Φ -guided pair having higher minimum Φ (0.9986 versus 0.7109). This counterexample was retained in the analysis rather than excluded. The second counterexample is `ibm_marrakesh` in Test 9, described in Subsection 3, where the RAW-selected pair had lower minimum Φ (0.9765) than the Φ -selected pair (0.9976) but achieved lower realized error under matched compiled circuit conditions. Together, these two counterexamples bound the Φ -based selection claim: composite Φ is often a useful selection signal, but calibration-proxy Φ does not dominate realized sampling error on every execution interval. Across the paid hardware suite, composite Φ outperformed raw T_2 in 3 of 4 matched comparisons (winning on `ibm_fez` in Test 8, and on `ibm_fez` and `ibm_torino` in Test 9; losing on `ibm_marrakesh` in Test 9) and lost the one random-baseline comparison in Test 6, with the two counterexamples marking the boundary of the calibration proxy.

4.10 Three-Layer Validation Summary

The 10 paid hardware tests summarized above constitute the primary empirical evidence. Two additional validation layers exercised the controller framework without consuming hardware execution credits. The simulated harness layer (January 11, 2026) triggered all four non-CONTINUE intervention actions under synthetic Φ sequences at $\Phi = 0.19$ across varying progress levels: CHECKPOINT-MIGRATE at cycle 8 (progress 70%), ABORT-RESTART at cycle 5 (progress 40%), CLASSICAL-FALLBACK at cycle 5 (progress 40%, 5 remaining qubits, 60 remaining gates), and CONTINUE-DEGRADED at cycle 11 (progress 91%). Each simulated scenario produced a structured audit record matching the decision specification. The IBM calibration harness layer connected to `ibm_fez` (156 qubits), retrieved live Φ values, and verified controller behavior under fault injection: the controller triggered CLASSICAL-FALLBACK at cycle 6 with 1 API call and 5 cached values, confirming the caching and throttling behavior.

Across the three layers, paid hardware tests directly validated CONTINUE (Test 2), CHECKPOINT-MIGRATE (Test 4), and ABORT-RESTART in both pre-submit and statistical forms (Tests 5 and 10). Test 7 separately validated the execution-time branching mechanism used to support mid-circuit intervention. CLASSICAL-FALLBACK was exercised in both the simulated harness and the telemetry-based IBM calibration harness under fault injection, while CONTINUE-DEGRADED was exercised in the simulated harness layer only. The scientific claim is bounded accordingly: the five-action taxonomy is instantiated across simulated, telemetry, and paid-hardware layers, with direct paid-hardware evidence for CONTINUE, CHECKPOINT-MIGRATE, and ABORT-RESTART, and supporting simulated or telemetry-layer evidence for CLASSICAL-FALLBACK and CONTINUE-DEGRADED. Total paid hardware credits consumed across the 10 paid tests were approximately 228 seconds of IBM Quantum Runtime time.

5 Discussion

The results provide affirmative evidence on the four questions raised in the Introduction, in the tested setting, while preserving documented boundaries.

On the first question, whether a controller can ingest Φ from live backend telemetry and use it to change realized hardware outcomes, the same-circuit comparison on `ibm_fez` is the clearest evidence. Swapping LOW- Φ qubits for HIGH- Φ qubits on an identical Deutsch-Jozsa circuit reduced error from 18.36% to 2.73%, an 85.1% relative reduction. Under matched conditions, composite Φ outperformed raw T_2 by 7.79 percentage points on `ibm_fez` (68.7% relative reduction) and by 6.37 percentage points aggregate across three backends. The composite metric adds measurable value over the strongest single-field calibration alternative, which addresses the obvious reviewer question of whether a single calibration field would suffice. The documented counterexamples bound this answer: Φ is a useful control variable, not an infallible oracle for realized error on every interval.

On the second question, whether statistical evidence can replace hardcoded fidelity thresholds, the Wilson confidence interval rule produced a principled trigger. A 3.82 percentage point separation between probe confidence intervals on `ibm_fez` was sufficient to trigger ABORT-RESTART and commit the full shot budget to the higher- Φ pool, which achieved 95.85% fidelity. On `ibm_marrakesh` and `ibm_torino`, overlapping intervals correctly held the controller at CONTINUE with the baseline pool. Intervention fired on 1 of 3 backends in the tested setting, which is the correct behavior when statistical separation is backend-dependent. This rule improves on a hardcoded fidelity cutoff for two reasons: it accounts for measurement uncertainty from finite shot counts, and it adapts more naturally to different backend quality regimes than a fixed threshold, without requiring a hardcoded fidelity value in the tested configuration.

On the third question, whether computational progress can be preserved when LOW- Φ qubits are detected after partial execution, the checkpoint-migrate pathway demonstrated a no-cloning-compatible solution. The classical representation of a measured outcome at a segmentation boundary is sufficient to resume a parameterized suffix computation on alternative qubits without storing an unknown quantum state. In the tested migration, the suffix on HIGH- Φ qubits achieved 82.15% fidelity after the prefix on LOW- Φ qubits produced a 92.58%-confidence checkpoint. The 82.15% suffix fidelity is lower than the 97.27% achieved by a HIGH- Φ pair on a shorter circuit in Test 2, reflecting the accumulated cost of segmentation and resume. The answer to the third question is therefore "useful progress preservation" rather than "lossless migration," and the applicability of this approach depends on whether the workflow admits a classically-measurable segmentation boundary.

On the fourth question, whether true mid-circuit intervention can be demonstrated on current IBM hardware, the dynamic-circuit branching test achieved 98.78% per-shot conditional consistency between probe outcome and branch marker across 4096 shots. This is direct evidence that mid-circuit measurement, classical conditional evaluation, and branch selection execute correctly on current hardware, rather than being a simulation-only capability. The per-shot marker-bit verification distinguishes this result from probe-only sentinel work that infers branch behavior from population statistics.

Taken together, the three-layer validation stack (simulated harness, telemetry harness, and 10 paid hardware tests on three backends) instantiates the five-action taxonomy with asymmetric evidence strength across actions. CONTINUE, CHECKPOINT-MIGRATE,

and ABORT-RESTART have direct paid-hardware evidence. CLASSICAL-FALLBACK has telemetry-layer and simulated evidence; CONTINUE-DEGRADED has simulated-layer evidence only. The appropriate scientific framing is that the framework is instantiated across the five actions and that paid-hardware evidence is layered, not uniform. The underlying open-loop limitation in current quantum execution workflows is the same problem that motivates pre-execution selection and routing; the contribution of this work is the during-execution control layer that complements those pre-execution approaches. Composition with post-compilation routing methods such as mapomatic (Nation and Treinish, 2023) is natural: a pre-execution routing decision can choose among available hardware targets, post-compilation layout optimization can select the best qubit assignment within the chosen target, and the closed-loop controller can intervene during execution when telemetry or probe measurements indicate degraded conditions.

6 Limitations

The results are bounded to three IBM Quantum backends (`ibm_fez`, `ibm_marrakesh`, `ibm_torino`) and to the paid test executions performed between January 15 and January 29, 2026. Whether the closed-loop control results generalize to other IBM backends, to other quantum hardware platforms such as those from Google, IonQ, or Rigetti, or to execution conditions outside the tested window is not established by this study.

Paid-hardware evidence does not cover all five intervention actions equally. CONTINUE, CHECKPOINT-MIGRATE, and ABORT-RESTART have direct paid-hardware tests. CLASSICAL-FALLBACK was exercised in the telemetry-based calibration harness and in the simulated harness but not in a paid circuit execution in this study. CONTINUE-DEGRADED was exercised in the simulated harness only. The scientific claim is bounded accordingly: the full five-action taxonomy is instantiated across layers, but the paid-hardware layer does not validate every action to the same degree.

Calibration-proxy Φ does not dominate realized sampling error on every execution interval. Two counterexamples were documented in the paid hardware suite: on `ibm_marrakesh` in Test 9, the RAW-selected pair had lower minimum Φ (0.9765) than the Φ -selected pair (0.9976) but achieved lower realized error under matched compiled circuit conditions; in Test 6 on `ibm_fez`, a seeded random pair outperformed the Φ -guided pair despite lower minimum Φ . The correct claim is that composite Φ outperformed raw T_2 in 3 of 4 matched comparisons (winning on `ibm_fez` and `ibm_torino`, losing on `ibm_marrakesh`) and lost the one random-baseline comparison on `ibm_fez`, not winning in every comparison. These counterexamples reflect genuine proxy disagreement intervals rather than calibration drift or implementation error.

The deep-circuit result in Test 3 is consistent with depth-induced error accumulation on marginal- Φ qubits under conditions that the periodic calibration snapshots did not surface as deteriorating. The experiment does not establish what fidelity a HIGH- Φ pair would have achieved at the same compiled depth of 3751, and it does not isolate mid-execution Φ collapse from depth-induced degradation on initially marginal qubits. The claim is bounded to realized depth degradation despite stable calibration snapshots in the tested window.

The Wilson confidence interval rule was tested at the 95% confidence level with 256-shot probes and a continue-baseline tie policy. Other confidence levels, probe sizes, and tie policies are permissible in the framework but were not evaluated in this study.

The thresholds used in the controller configuration (intervention threshold $\Phi_c = 0.25$, coupling constant $\alpha = 0.1$, near-completion progress 85%, migration-eligibility progress 50%, CLASSICAL-FALLBACK bounds of 25 qubits and 1000 gates, confirmation window 2 cycles, cooldown 3 cycles, maximum 5 interventions) are configurable parameters. The values used correspond to the reference configuration for the tested experiments and may warrant adjustment under different deployment characteristics, circuit families, or cost models.

Pre-submit ABORT-RESTART in Test 5 does not measure the realized fidelity that the avoided LOW- Φ pair would have produced. The supported claim is preventive resource saving rather than a measured completed-run counterfactual. The mid-circuit branching test in Test 7 produced 25.27% overall fidelity by design; the test verified branching mechanics rather than algorithmic fidelity, and the 98.78% consistency figure refers to probe-marker agreement rather than to circuit correctness.

Reproducibility was not evaluated exhaustively across all 10 paid tests. Run-to-run variation between independent replications of similar comparisons (for example, the approximately 0.05 to 0.29 percentage point differences between Test 8 and Test 9 `ibm_fez` results on the same qubit pairs) is consistent with finite-sample measurement uncertainty at 4096 shots but is not systematically characterized here.

7 Conclusion

We investigated whether a training-free thermodynamic stability metric $\Phi = I \times \rho - \alpha \times S$, computed from IBM Quantum calibration or telemetry data, can drive closed-loop intervention decisions during execution-associated monitoring windows on real hardware. Across 10 paid circuit tests on three IBM backends (`ibm_fez`, `ibm_marrakesh`, `ibm_torino`), together with simulated and telemetry-based harness validation, we evaluate a closed-loop controller that monitors Φ during an execution-associated window and supports five intervention actions: CONTINUE, CONTINUE-DEGRADED, CLASSICAL-FALLBACK, CHECKPOINT-MIGRATE, and ABORT-RESTART. The main findings are:

1. Swapping LOW- Φ qubits for HIGH- Φ qubits on an identical Deutsch-Jozsa circuit on `ibm_fez` reduced error from 18.36% to 2.73%, an 85.1% relative error reduction, establishing closed-loop Φ -driven selection as an actionable control decision in the tested setting.
2. Under matched conditions, composite Φ outperformed raw T_2 by 7.79 percentage points on `ibm_fez` and by 6.37 percentage points aggregate across three backends. Composite Φ won 3 of 4 matched raw- T_2 comparisons with a documented counterexample on `ibm_marrakesh`, and lost the one random-baseline comparison in Test 6.
3. A Wilson confidence interval non-overlap rule triggered ABORT-RESTART on 1 of 3 backends in the tested setting, with a 3.82 percentage point separation sufficient to drive the intervention; the full Φ run on that backend achieved 95.85% fidelity.
4. A classical representation of a measured outcome at a segmentation boundary enabled suffix migration from LOW- Φ to HIGH- Φ qubits with 82.15% final fidelity, without storing an unknown quantum state.

5. Mid-circuit dynamic branching on IBM hardware achieved 98.78% per-shot conditional consistency between probe outcome and branch marker across 4096 shots, establishing that mid-circuit intervention is a working hardware capability rather than a simulation-only construct.

The paid-hardware evidence layer is asymmetric across the five intervention actions. CONTINUE, CHECKPOINT-MIGRATE, and ABORT-RESTART are validated directly in paid circuit execution; CLASSICAL-FALLBACK is validated in telemetry and simulated layers; CONTINUE-DEGRADED is validated in the simulated harness only. Two documented counterexamples bound the Φ -based selection claim: on `ibm_marrakesh` in Test 9 the RAW-selected pair outperformed the Φ -selected pair despite lower minimum Φ on the winning pair, and in Test 6 on `ibm_fez` a seeded random pair outperformed the Φ -guided pair. These results support a closed-loop intervention framework in which Φ operates as a control variable during execution-associated monitoring, complementing pre-execution selection and routing work and composing naturally with post-compilation layout optimization and hybrid allocation methods.

Code Availability

Code, repository documentation, test plans, and paid-hardware results summaries are available at: <https://github.com/Wise314/phi-controller-quantum>.

Acknowledgments

IBM Quantum hardware accessed via IBM Quantum Network.

Patent Disclosures

The methods described in this paper are the subject of U.S. Provisional Patent Application No. 63/973,723 (filed February 2, 2026). No license to implement or commercialize the described methods is granted by this publication. All rights reserved.

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